

Unit 02: Installation Guide

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Objectives

After studying this unit, you will be able to

- Understand the booting process
- Understand the installation process of Linux
- Understand the partitioning of hard drives
- Understand the file system types
- Understand how to log in the system

Introduction: An operating system (OS) is a system software that manages computer hardware, software resources, and provides common services for computer programs. Booting is a bootstrapping process that starts operating systems when the user turns on a computer system. A boot sequence is the set of operations the computer performs when it is switched on that load an operating system.

2.1 Booting sequence:

The booting sequence follows several steps. These are:

- Turn on the system
- CPU jump to address of BIOS
- BIOS runs POST (Power-On Self-Test)
- Find bootable devices
- Loads and execute boot sector form MBR
- Load OS

The first and foremost task is to turn on the system. So that the other processes can start. Rest of the process includes jumping of CPU to the address of CPU, BIOS runs POST, finding of bootable devices, loads and execute boot sector from MBR and loading of operating system.



BIOS (Basic Input/ Output System)

BIOS refer to the software code run by a computer when first powered on. It identifies your computer's hardware, configures it, tests it, and connects it to the operating system for further instruction. This is called the boot process. The primary function of BIOS is code program embedded on a chip that recognizes and controls various devices that make up the computer.

MBR(Master Boot Record)

OS is booted from a hard disk, where the Master Boot Record (MBR) contains the primary boot loader. The MBR is a 512-byte sector, located in the first sector on the disk (sector 1 of cylinder 0, head 0). After the MBR is loaded into RAM, the BIOS yields control to it.

Boot loader

Boot loader could be more adeptly called the kernel loader. The task at this stage is to load the Linux kernel. GRUB and LILO are the most popular Linux boot loader. Examples of boot loaders are:



Example: GRUB, LILO, GRUB2WIN, BOOTCAMP, BOOTKEY, NTLDR and Syslinux

GRUB:

GRUB stands for GRand Unified Bootloader. It is an operating system independent boot loader. It is a multiboot software packet from GNU. It has a flexible command line interface. It has file system access. It supports multiple executable formats. It supports diskless system.

LILO: LInux Loader

This boot loader does not depend on a specific file system. It can boot from hard-disk and floppy

Task of kernel

The kernel helps in process management, memory management. The device management is also one of the tasks of kernel. The system calls are also handled by kernel.

Init process

The first thing the kernel does is to execute init program. Init is the root/parent of all processes executing on Linux. The first processes that init starts is a script `/etc/rc.d/rc.sysinit`. Based on the appropriate run-level, scripts are executed to start various processes to run the system and make it functional. The init process is identified by process id "1". Init is responsible for starting system processes as defined in the `/etc/inittab` file. Upon shutdown, init controls the sequence and processes for shutdown.

Runlevels

A run-level is a software configuration of the system which allows only a selected group of processes to exist. Init can be in one of seven run-levels: 0-6.

Runlevel	ScriptsDirectory (RedHat/Fedora Core)	State
0	<code>/etc/rc.d/rc0.d/</code>	shutdown/halt system
1	<code>/etc/rc.d/rc1.d/</code>	Single user mode
2	<code>/etc/rc.d/rc2.d/</code>	Multiuser with no network services exported
3	<code>/etc/rc.d/rc3.d/</code>	Default text/console only start. Full multiuser
4	<code>/etc/rc.d/rc4.d/</code>	Reserved for local use. Also X-windows (Slackware/BSD)
5	<code>/etc/rc.d/rc5.d/</code>	XDM X-windows GUI mode (Redhat/System V)
6	<code>/etc/rc.d/rc6.d/</code>	Reboot
s or S		Single user/Maintenance mode (Slackware)
M		Multiuser mode (Slackware)

2.2 Download Linux

- To install Red Hat, you will need to download the ISO images (CD Images) of the installation CD-ROMs from <http://fedora.redhat.com>
- Download the i386 images for 32 Intel Processors, PPC images for Apple Macintosh and x86_64 for 64-bit AMD Processors.

Here the installation of Linux distribution is shown under VMWare. It can be installed as an individual operating system as well. Generally, when we buy a new system, the seller gives us the Windows operating system by default in that.

2.3 Installation of Linux

New Virtual Machine Wizard:

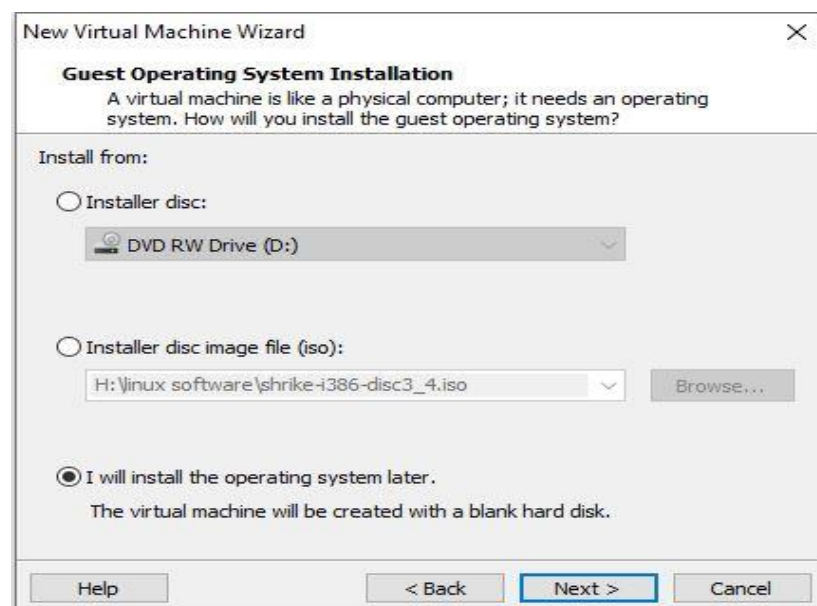
After the installation of VMWare in the system, the New Virtual Machine Wizard will open. It asks for the type of configuration you want to go with: typical or custom. The typical configuration is the recommended one and custom is the configuration with the advanced options. Choose the appropriate option and go to next step.



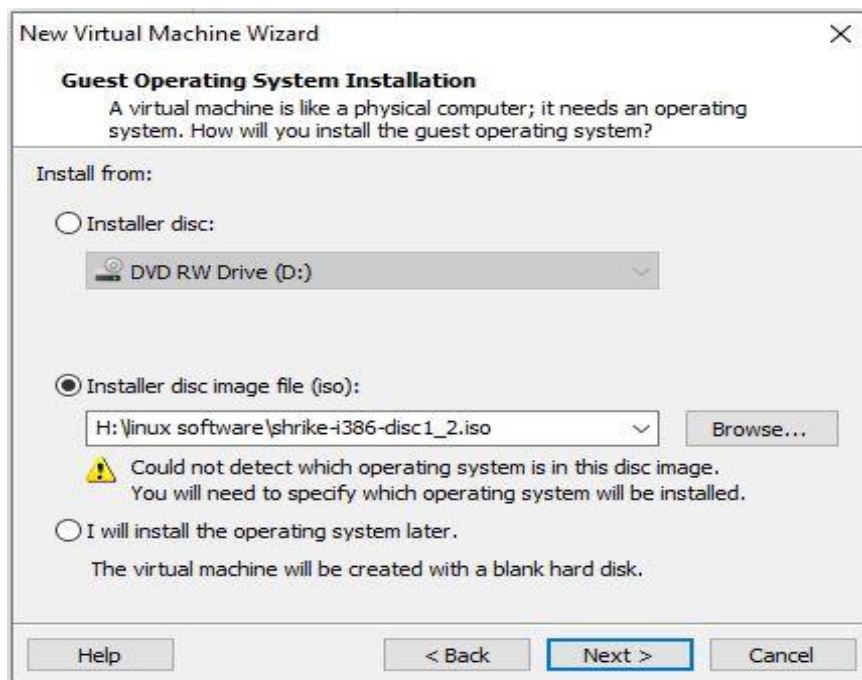
Installation of operating system

In this step, the guest operating system will be installed. Here the available options are

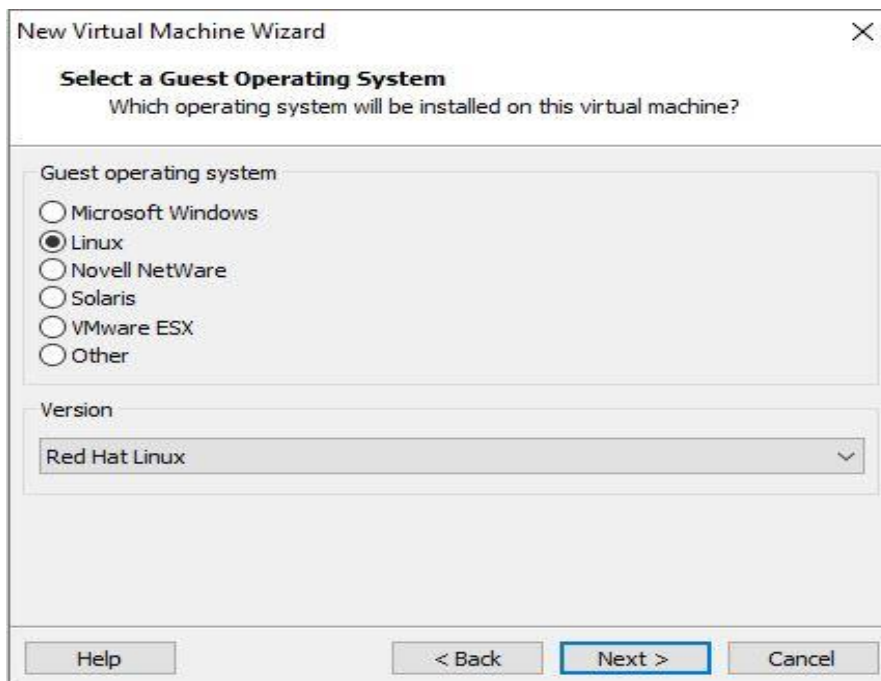
- Installer disc
- Installer disc image file (iso)
- Later installation.



So out of these three options, choose the option depending upon the availability of the operating system. Here the installation using iso images is shown. The whole operating system is divided into three iso images.



After selecting the first iso image, choose which guest operating system you want to install. Here the Linux is chosen for installation on the virtual machine. The version of the operating system will also be selected here. Once it is done, click on Next to go further.



Write the name of the virtual machine and choose the location where you want to have all of its files. Once this is done, click on next to go further.

The screenshot shows the 'New Virtual Machine Wizard' dialog box with the title 'Name the Virtual Machine'. Below the title is the question 'What name would you like to use for this virtual machine?'. There are two input fields: 'Virtual machine name:' with the text 'Red Hat Linux (18)' and 'Location:' with the text 'C:\Users\Divya Singh\Documents\Virtual Machines\Red Hat Linu:'. To the right of the 'Location' field is a 'Browse...' button. Below the input fields is the text 'The default location can be changed at Edit > Preferences.' At the bottom of the dialog are three buttons: '< Back', 'Next >' (which is highlighted with a blue border), and 'Cancel'.

Specify disk capacity

When the name and location is specified, next task is to specify the disk capacity. By default, the recommended maximum disk size is 8 GB. We can increase or decrease it as per our requirements. Then it will ask whether you want to store virtual disk as a single file or into multiple files. Choose the appropriate option and go further.

The screenshot shows the 'New Virtual Machine Wizard' dialog box with the title 'Specify Disk Capacity'. Below the title is the question 'How large do you want this disk to be?'. There is a text block explaining that the virtual machine's hard disk is stored as one or more files on the host computer's physical disk. Below this is a 'Maximum disk size (GB):' label followed by a text box containing '8.0' and a spinner button. Below that is the text 'Recommended size for Red Hat Linux: 8 GB'. There are two radio button options: 'Store virtual disk as a single file' and 'Split virtual disk into multiple files' (which is selected). Below the radio buttons is a text block explaining that splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks. At the bottom of the dialog are four buttons: 'Help', '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

After this, there are two options. One is to power on this virtual machine and other is to edit virtual machine settings. If you have done any mistake while setting the machine, then edit those settings. Otherwise turn on the power machine.



When you power on the virtual machine, the next screen will be:

```

- To install or upgrade Red Hat Linux in graphical mode,
  press the <ENTER> key.

- To install or upgrade Red Hat Linux in text mode, type:
  linux text <ENTER>.

- Use the function keys listed below for more information.

[F1-Main] [F2-Options] [F3-General] [F4-Kernel] [F5-Rescue]
boot: _

```

When you press the ENTER key, the next screen will be:

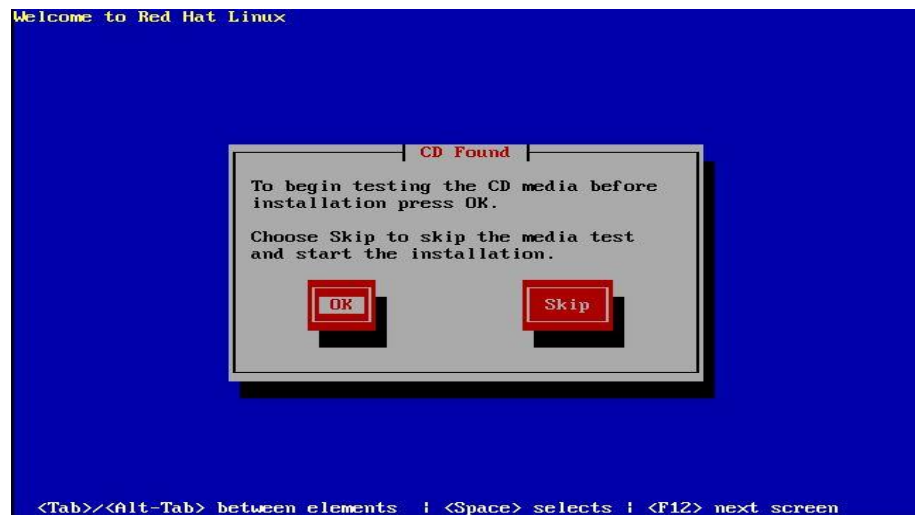
```

hid-core.c: USB HID support drivers
mice: PS/2 mouse device common for all mice
md: md driver 0.90.0 MAX_MD_DEVS=256, MD_SB_DISKS=27
md: Autodetecting RAID arrays.
md: autorun ...
md: ... autorun DONE.
NET4: Linux TCP/IP 1.0 for NET4.0
IP Protocols: ICMP, UDP, TCP
IP: routing cache hash table of 2048 buckets, 16Kbytes
TCP: Hash tables configured (established 16384 bind 32768)
NET4: Unix domain sockets 1.0/SMP for Linux NET4.0.
RAMDISK: Compressed image found at block 0
Freeing initrd memory: 2640k freed
EXT2-fs warning: checktime reached, running e2fsck is recommended
UFS: Mounted root (ext2 filesystem).
Greetings.
Red Hat install init version 9.0 starting
mounting /proc filesystem... done
mounting /dev/pts (unix98 pty) filesystem... done
checking for NFS root filesystem...no
trying to remount root filesystem read write... done
checking for writeable /tmp... yes
running install...
running /sbin/loader

```

Testing the CD Media

If you are using the CD media for the first time, then it is advised to test it before the installation process. If the same media is used for installation earlier, then it can be skipped.



Running Anaconda

The Anaconda starts running. It is best to install Anaconda for the local user, which does not require administrator permissions and is the most robust type of installation. However, if you need to, you can install Anaconda system wide, which does require administrator permissions.

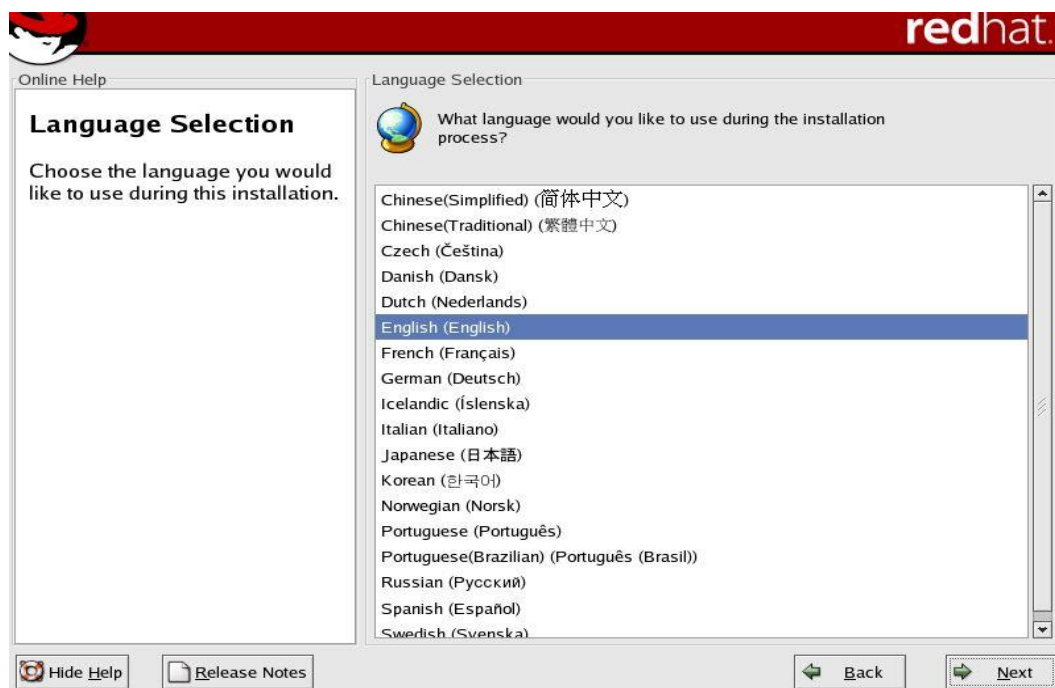


The process starts with the screen "Welcome to Red Hat Linux". Click on Next for further process.



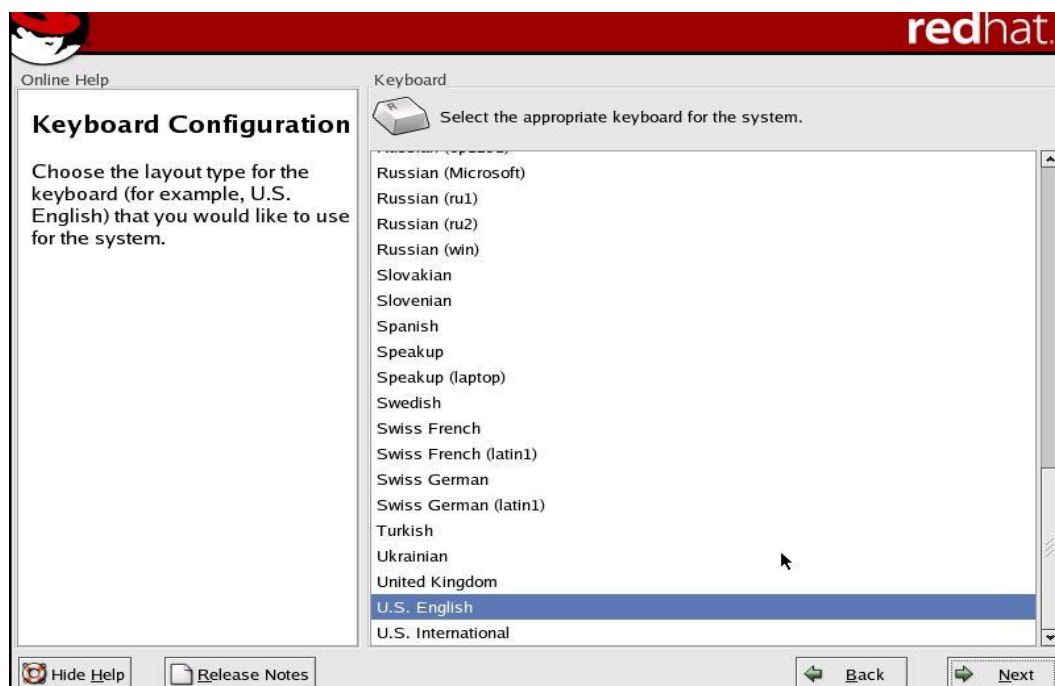
Language Selection

The next screen is for language selection. Choose the language that you want to use during installation process. After selection of language, click on next.



Keyboard Configuration

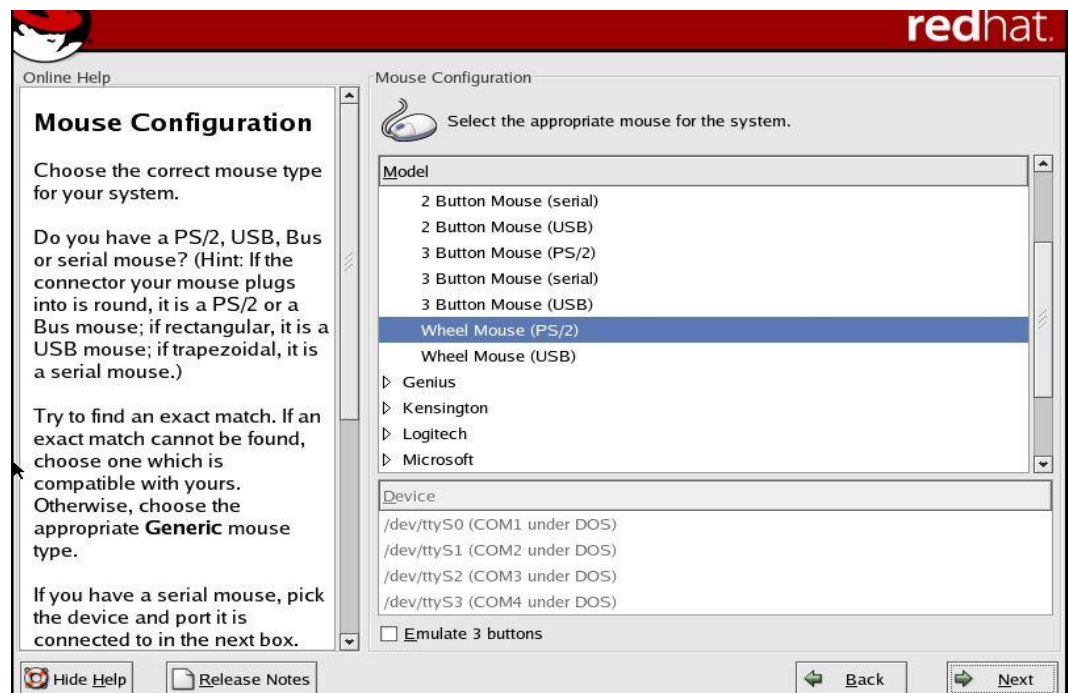
The screen is for choosing of layout type of keyboard that we want to use for the system. After chosen, click on next.



Linux and Shell Scripting

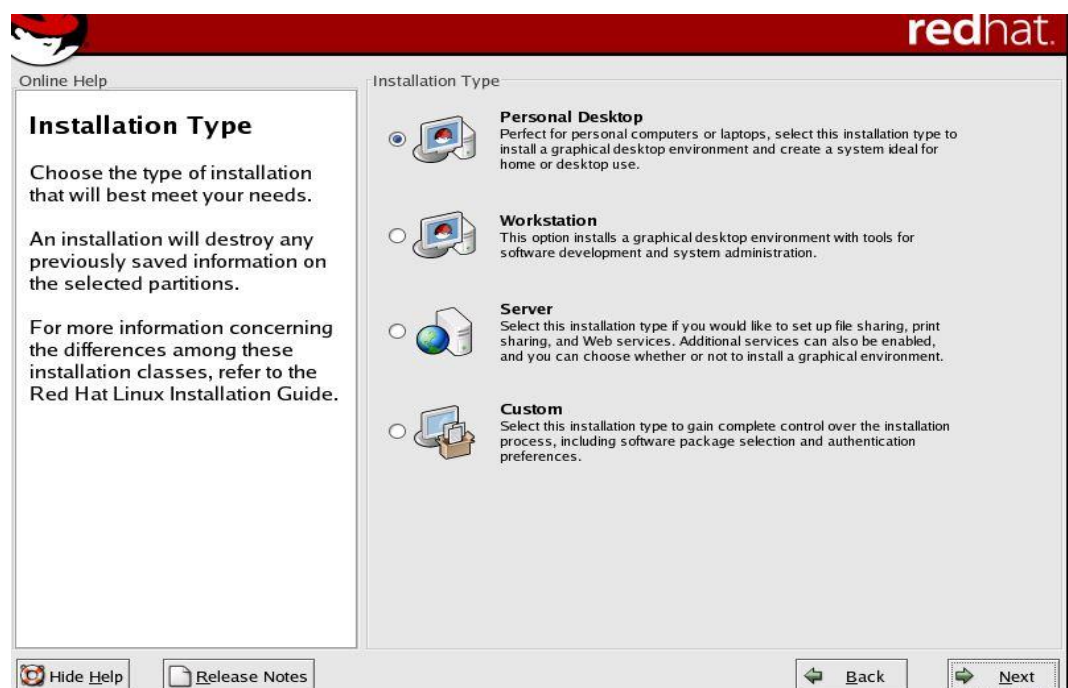
Mouse Configuration

The screen is for mouse configuration. By default, it will choose the appropriate option. So, it is advised to go with the recommended settings. Click on next.



Installation Type

Next is to choose the installation type which best meet your needs. The installation type available are: Personal Desktop, workstation, server and custom. Choose the appropriate option and click on next.



Disk Partitioning Setup

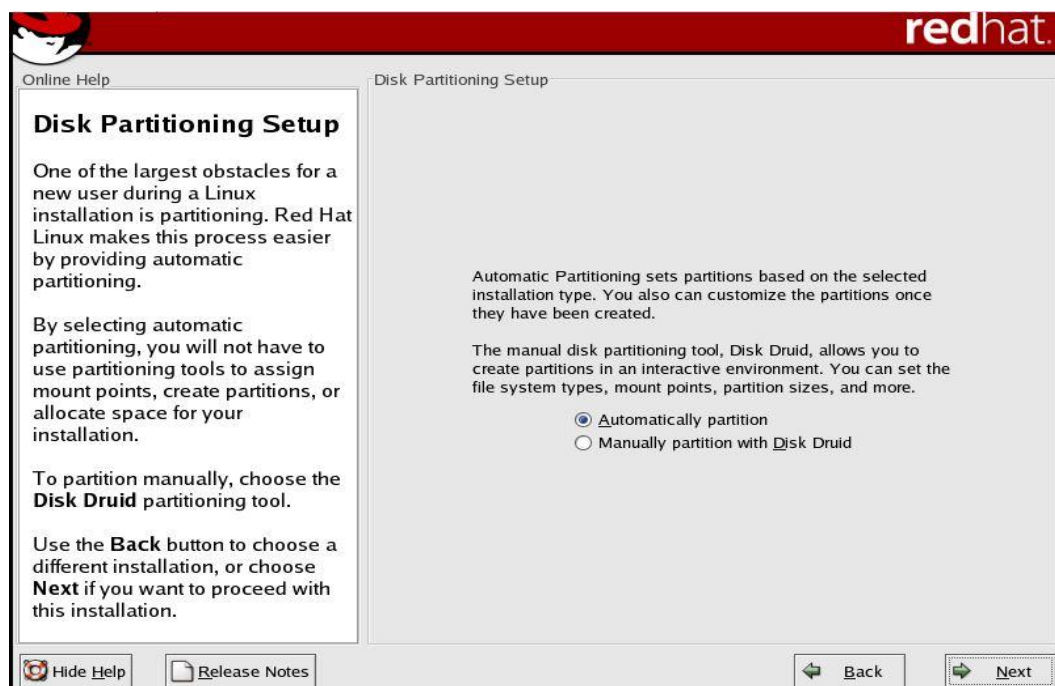
Partitioning is a means to divide a single hard drive into many logical drives. A partition is a contiguous set of blocks on a drive that are treated as an independent disk. A partition table is an index that relates sections of the hard drive to partitions.

The task is now to partition the drive. It can be done in two ways: automatically partition and manually partition with disk druid. The automatically partition is dependent upon the selected installation type. The manual partition uses the tool Disk Druid for partitioning. The partition fields are:

- **Device:** This field displays the partition's device name.
- **Start:** This field shows the sector on your hard drive where the partition begins.
- **End:** This field shows the sector on your hard drive where the partition ends.
- **Size:** This field shows the partition's size (in MB).
- **Type:** This field shows the partition's type (for example, ext2, ext3, or vfat).
- **Mount Point:** A mount point is the location within the directory hierarchy at which a volume exists; the volume is "mounted" at this location. This field indicates where the partition will be mounted.

The file system types are:

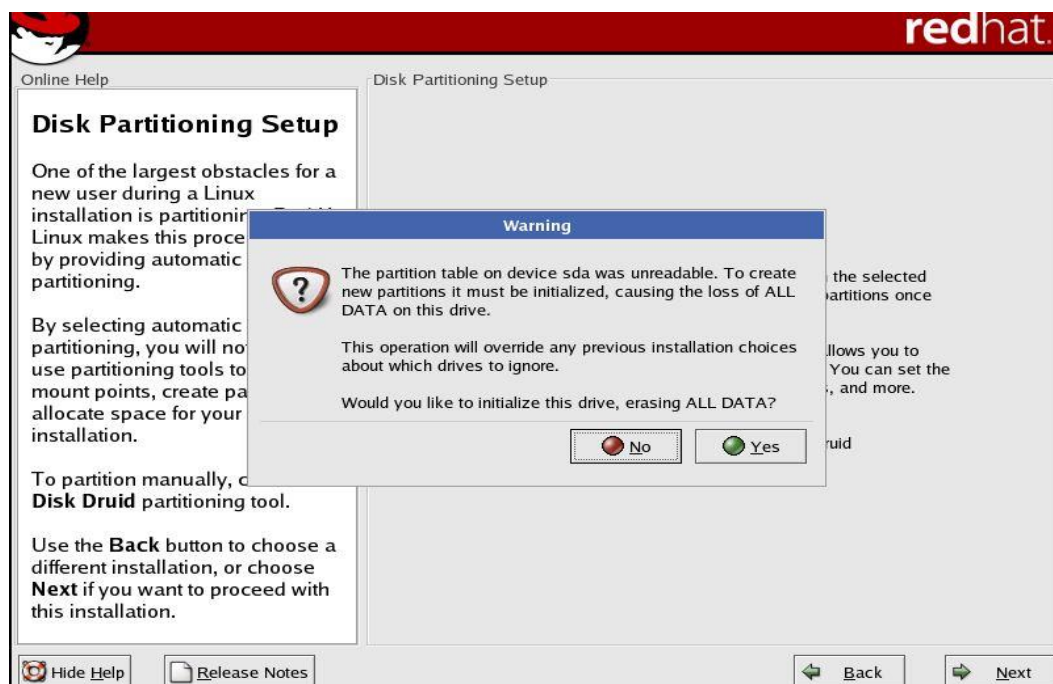
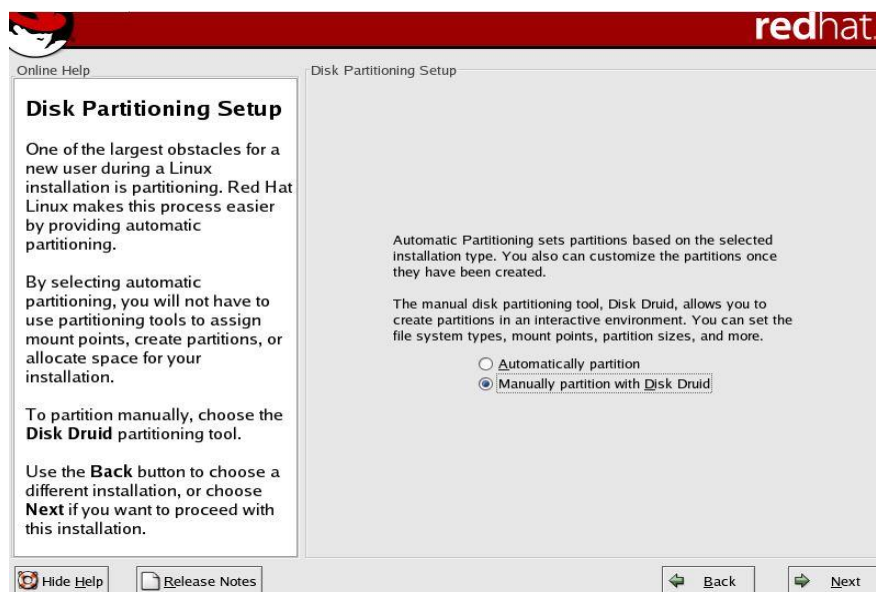
- **ext2** — An ext2 filesystem supports standard Unix file types (regular files, directories, symbolic links, etc). It provides the ability to assign long file names, up to 255 characters. Versions prior to Red Hat Linux 7.2 used ext2 file systems by default.
- **ext3** — The ext3 filesystem is based on the ext2 filesystem and has one main advantage — journaling. Using a journaling filesystem reduces time spent recovering a filesystem after a crash as there is no need to fsck the filesystem.
- **swap** — Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.
- **vfat** — The VFAT filesystem is a Linux filesystem that is compatible with Windows 95/NT long filenames on the FAT filesystem.

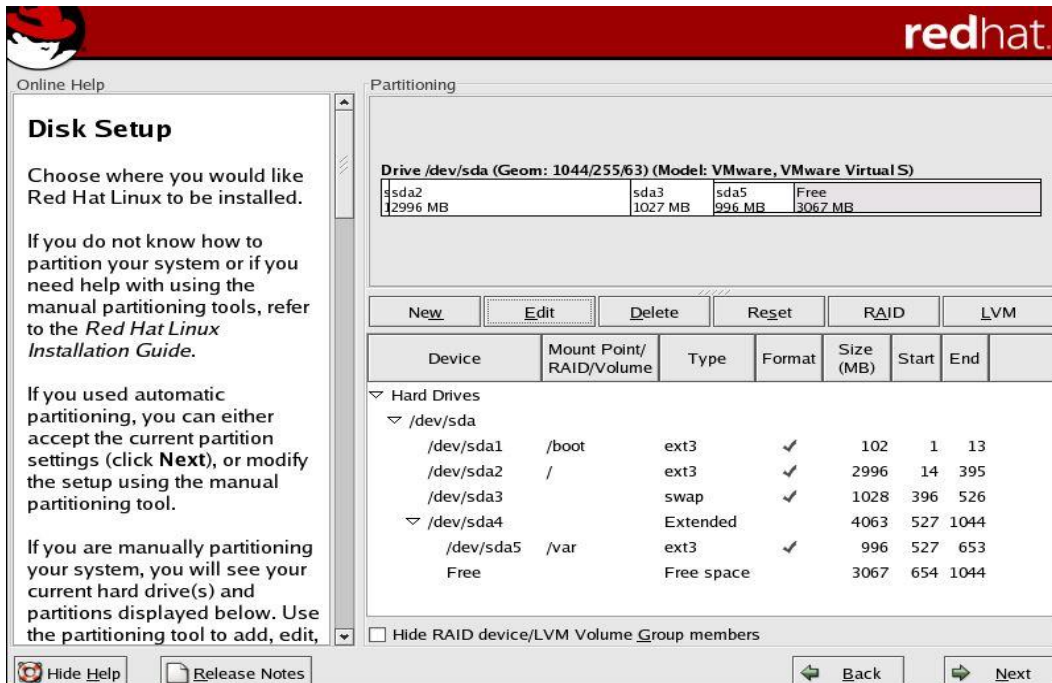


Recommended Partitioning Scheme

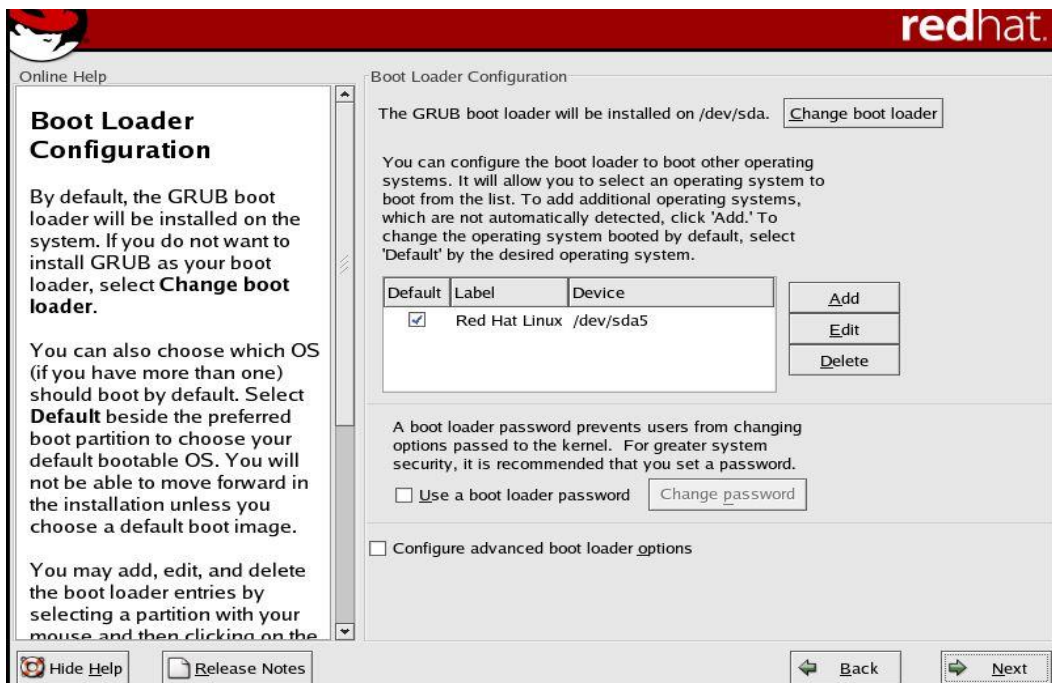
- Unless you have a reason for doing otherwise, it is recommended that you create the following partitions:
- /boot partition – contains kernel images and grub configuration and commands
- / partition
- /var partition
- Any other partition based on application (e.g /usr/local for squid)
- swap partition – swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing. The size of your swap partition should be equal to twice your computer's RAM.

Here we are going with manual partition in which disk druid tool helps.

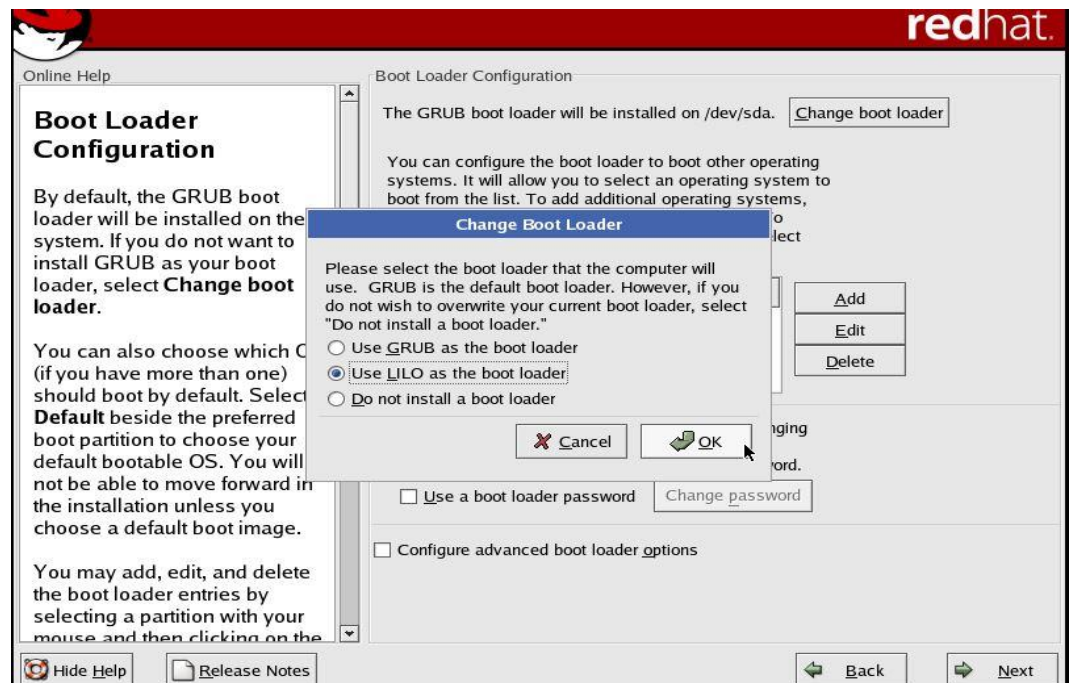
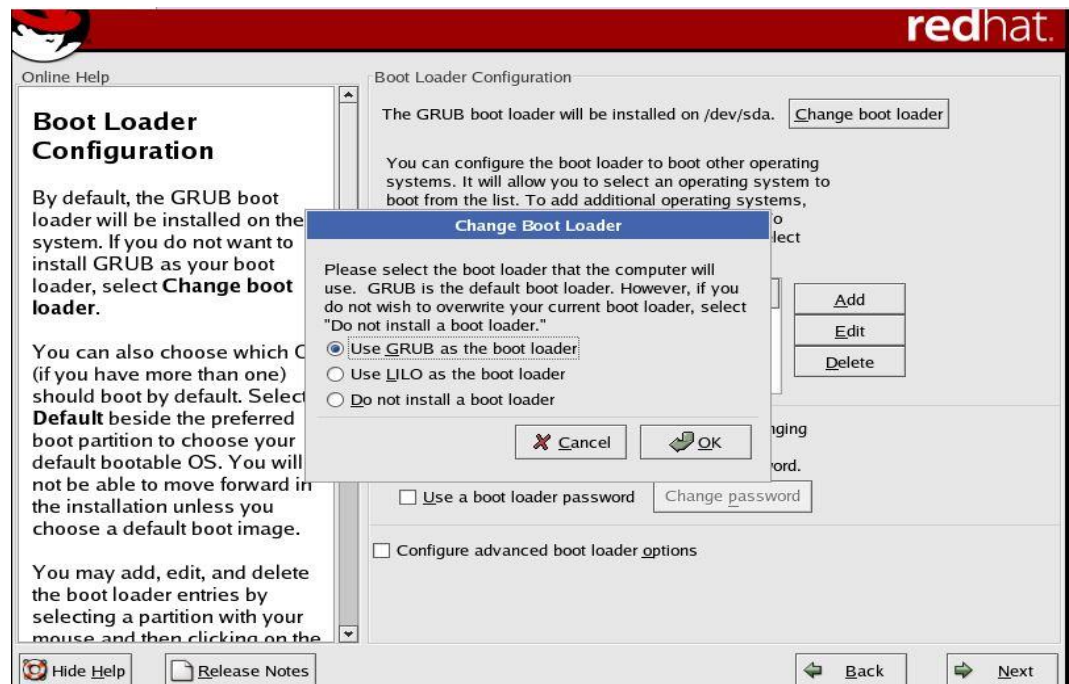




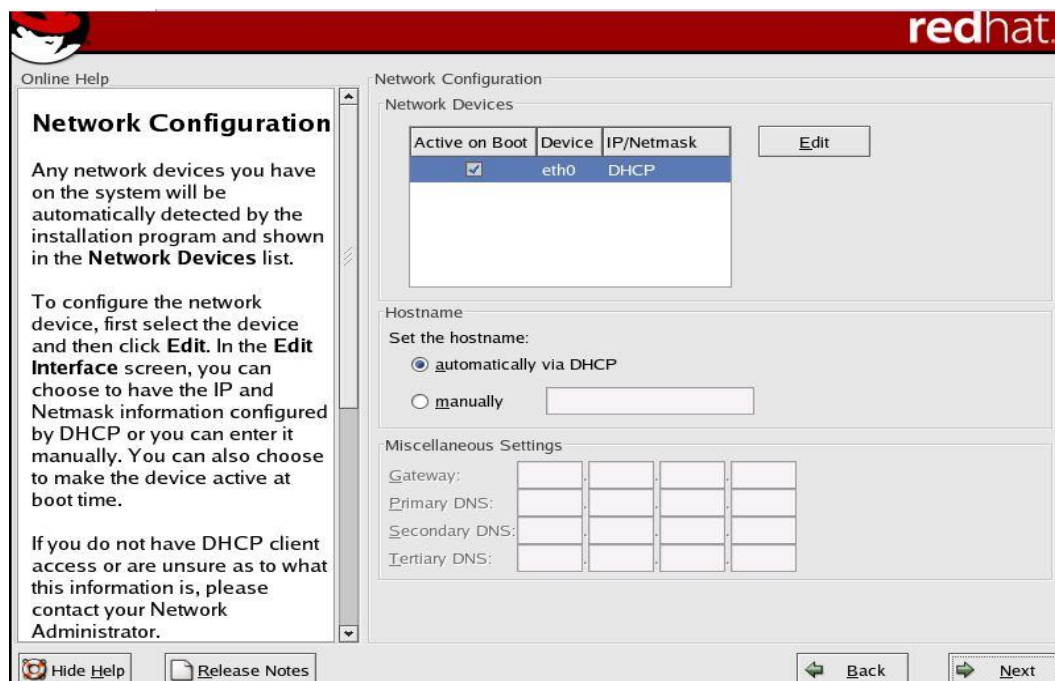
Boot Loader Configuration



Change Boot Loader



Network Configuration



Online Help

Network Configuration

Any network devices you have on the system will be automatically detected by the installation program and shown in the **Network Devices** list.

To configure the network device, first select the device and then click **Edit**. In the **Edit Interface** screen, you can choose to have the IP and Netmask information configured by DHCP or you can enter it manually. You can also choose to make the device active at boot time.

If you do not have DHCP client access or are unsure as to what this information is, please contact your Network Administrator.

Hide Help | Release Notes

Back | Next

redhat.

Network Configuration

Network Devices

Active on Boot	Device	IP/Netmask
☒	eth0	DHCP

Edit

Hostname

Set the hostname:

☒ automatically via DHCP

☐ manually

Miscellaneous Settings

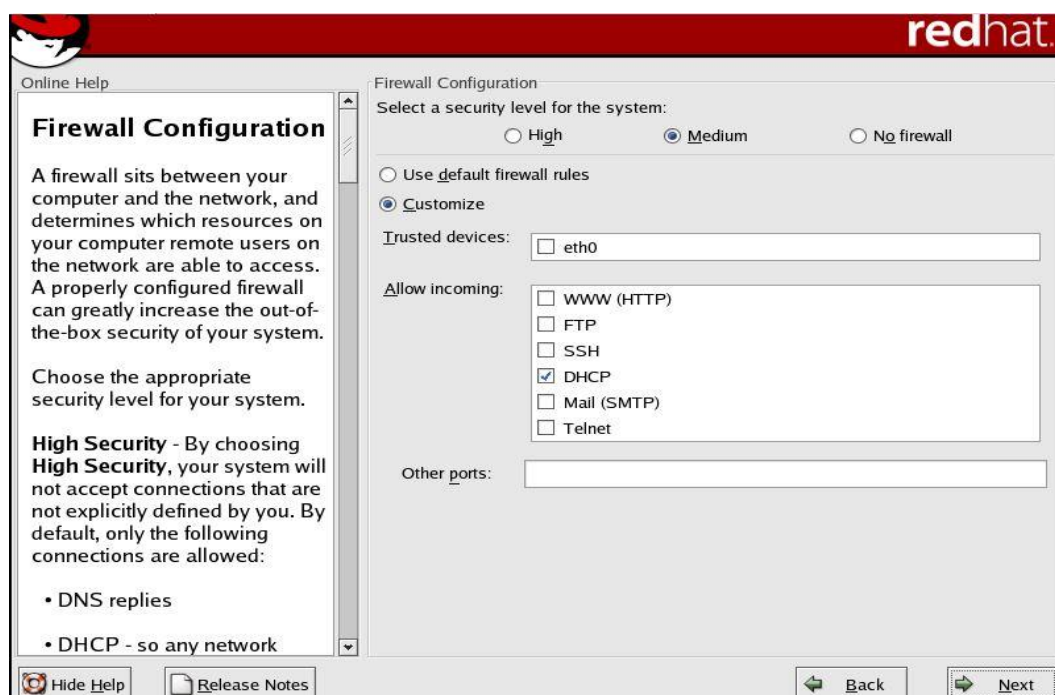
Gateway:

Primary DNS:

Secondary DNS:

Tertiary DNS:

Firewall Configuration



Online Help

Firewall Configuration

A firewall sits between your computer and the network, and determines which resources on your computer remote users on the network are able to access. A properly configured firewall can greatly increase the out-of-the-box security of your system.

Choose the appropriate security level for your system.

High Security - By choosing **High Security**, your system will not accept connections that are not explicitly defined by you. By default, only the following connections are allowed:

- DNS replies
- DHCP - so any network

Hide Help | Release Notes

Back | Next

redhat.

Firewall Configuration

Select a security level for the system:

☐ High ☒ Medium ☐ No firewall

☐ Use default firewall rules

☒ Customize

Trusted devices:

Allow incoming:

☐ WWW (HTTP)

☐ FTP

☐ SSH

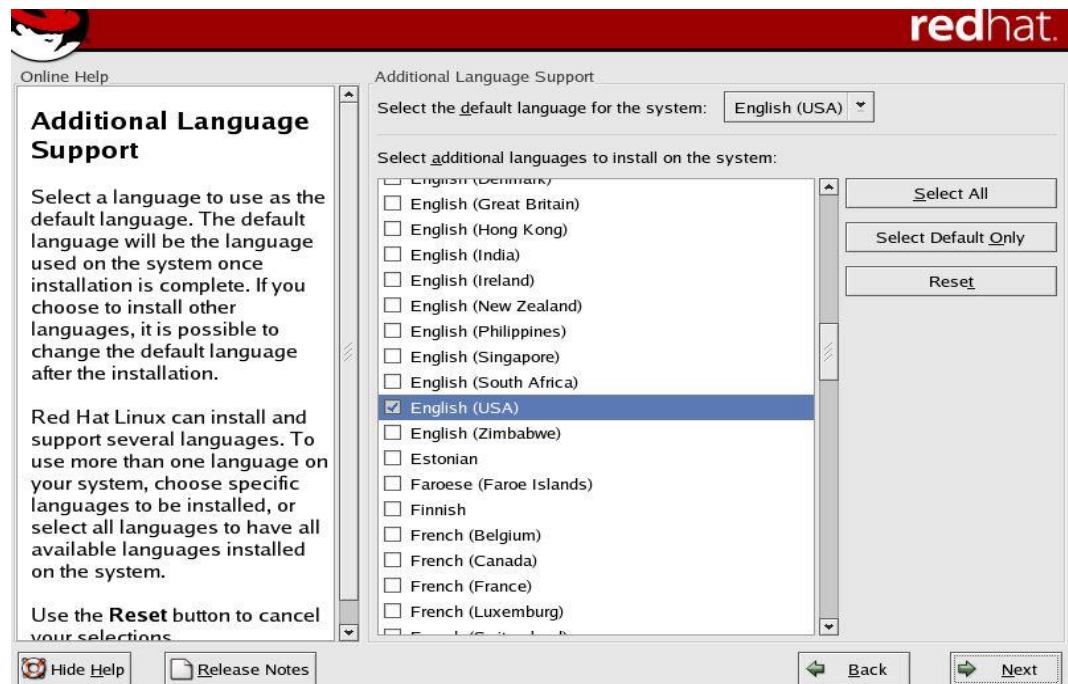
☒ DHCP

☐ Mail (SMTP)

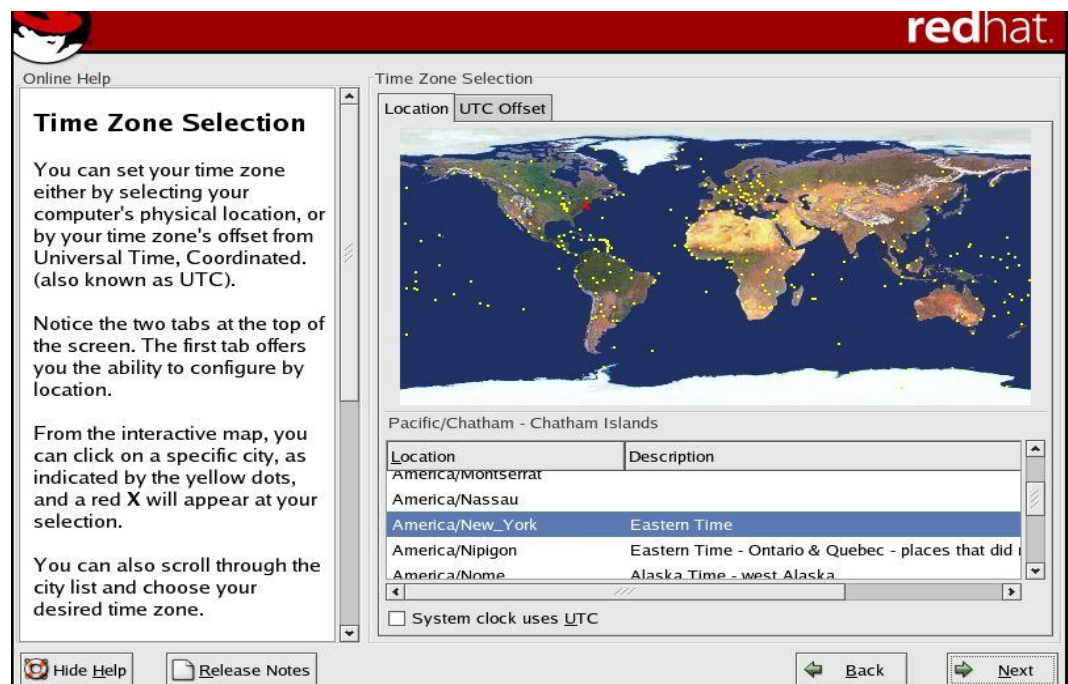
☐ Telnet

Other ports:

Additional Language Support



Time Zone Selection



Set Root Password



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Online Help

Set Root Password

Use the root account *only* for administration. Once the installation has been completed, create a non-root account for your general use and `su -` to gain root access when you need to fix something quickly. These basic rules will minimize the chances of a typo or incorrect command doing damage to your system.

Set Root Password

Enter the root (administrator) password for the system.

Root Password:

Confirm:

 Hide Help

 Release Notes

 Back

 Next

Provide Root Password



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Online Help

Set Root Password

Use the root account *only* for administration. Once the installation has been completed, create a non-root account for your general use and `su -` to gain root access when you need to fix something quickly. These basic rules will minimize the chances of a typo or incorrect command doing damage to your system.

Set Root Password

Enter the root (administrator) password for the system.

Root Password:

Confirm:

Root password accepted.

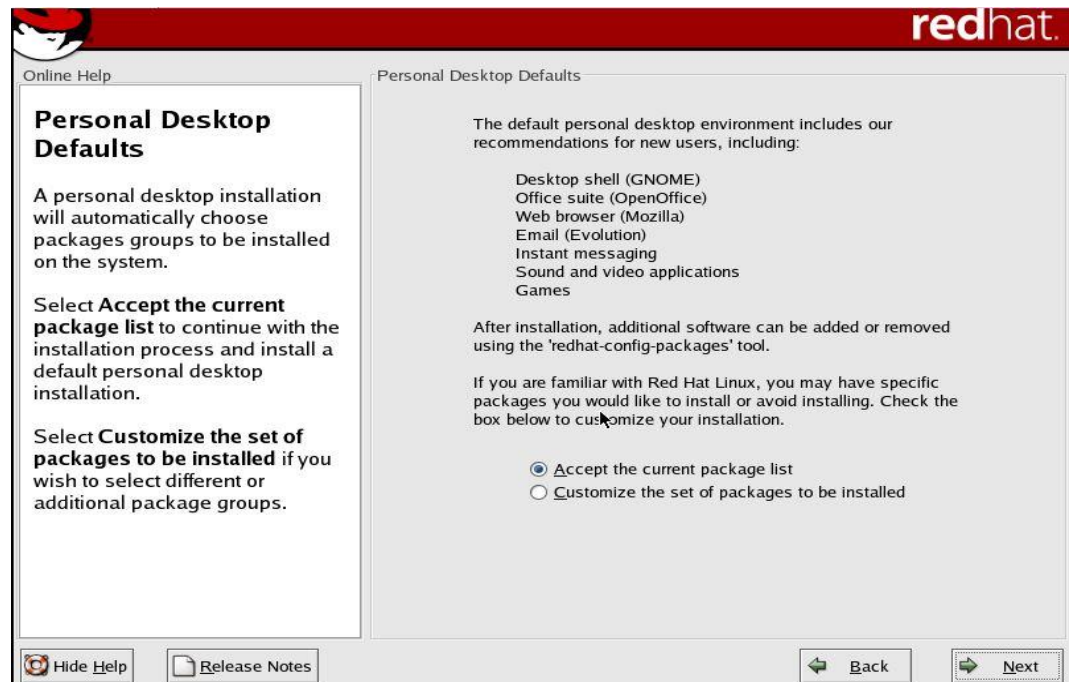
 Hide Help

 Release Notes

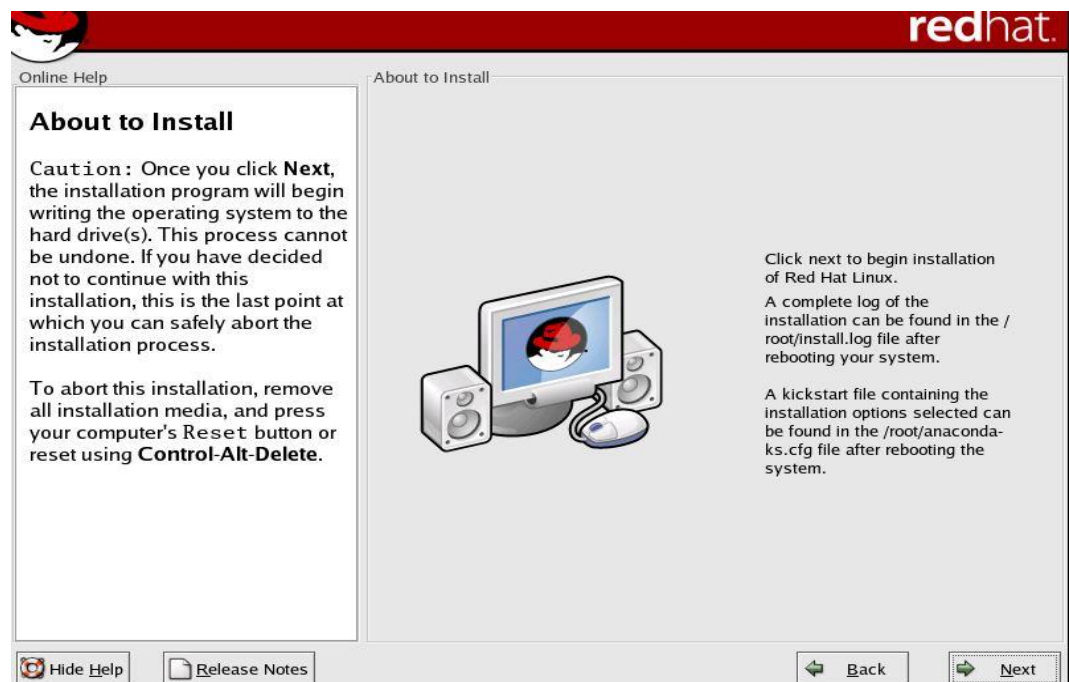
 Back

 Next

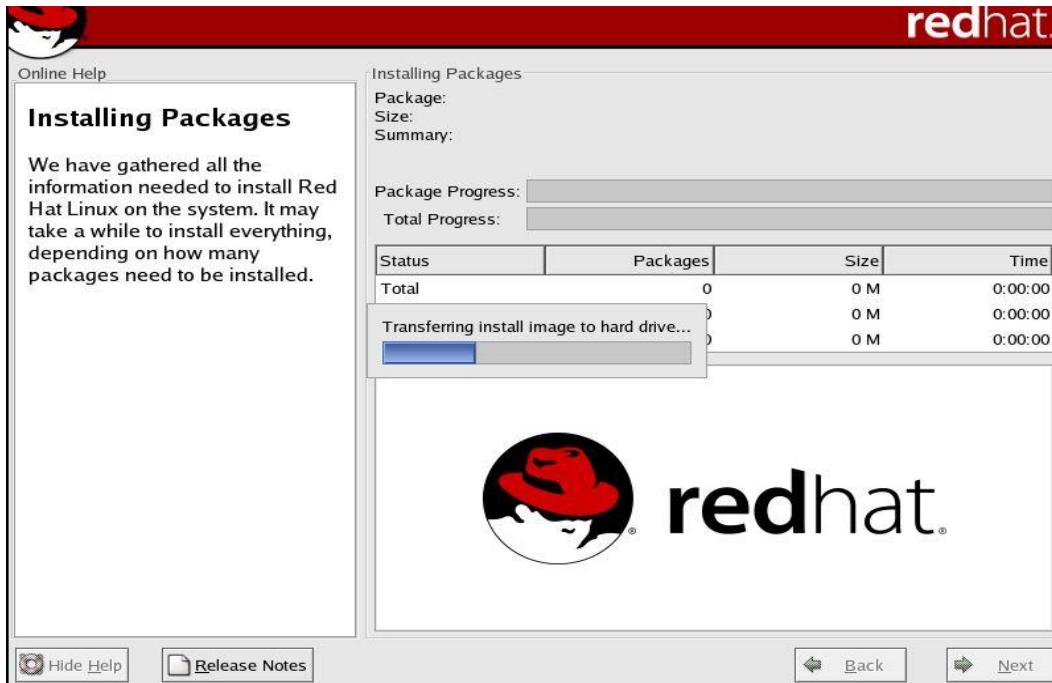
Personal Desktop Defaults (Packages)



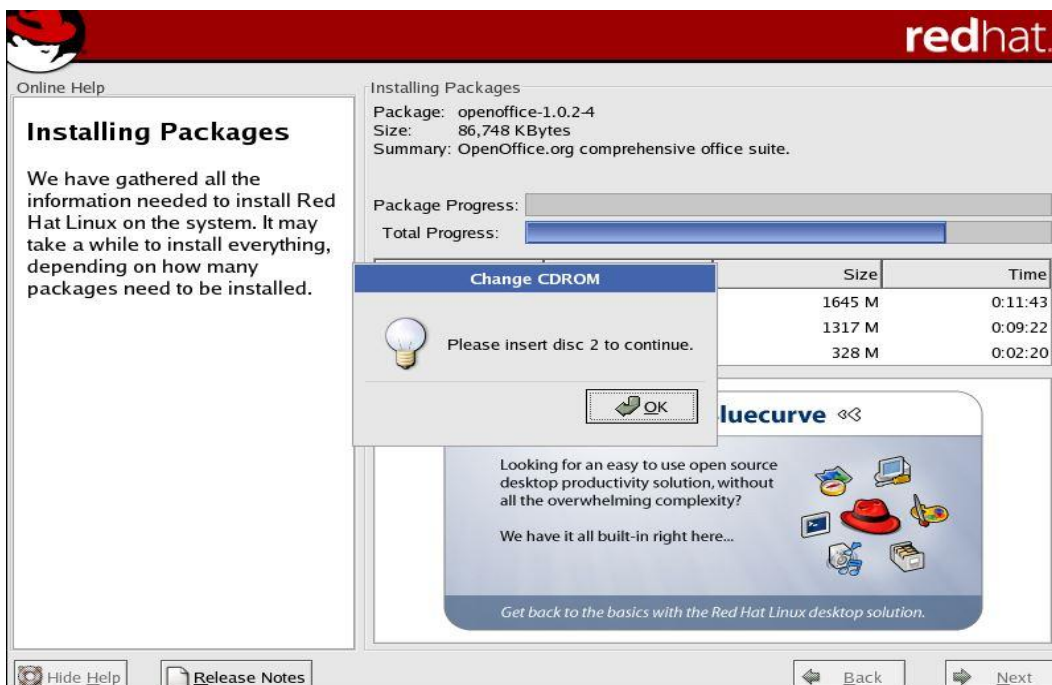
About to Install



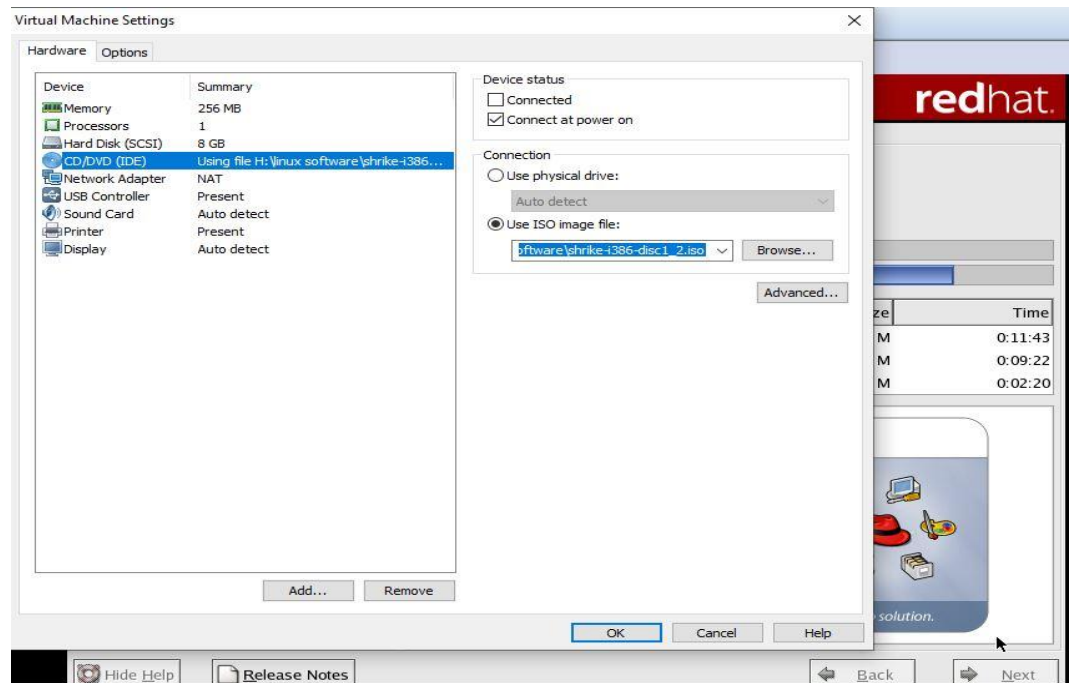
Installing Packages



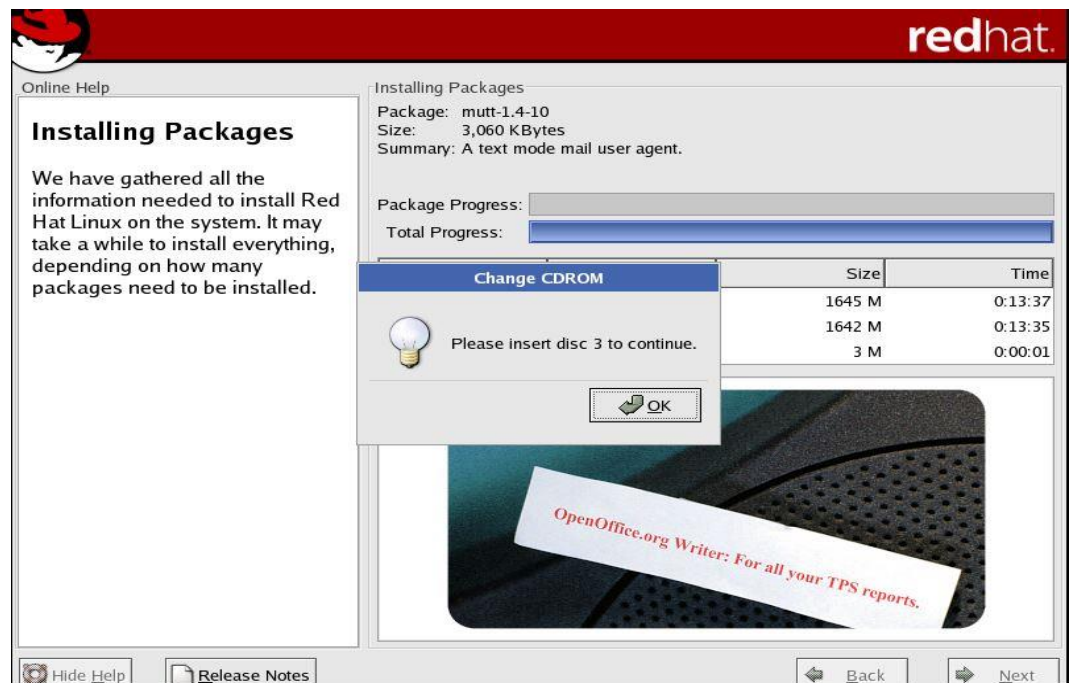
Insert Disc 2



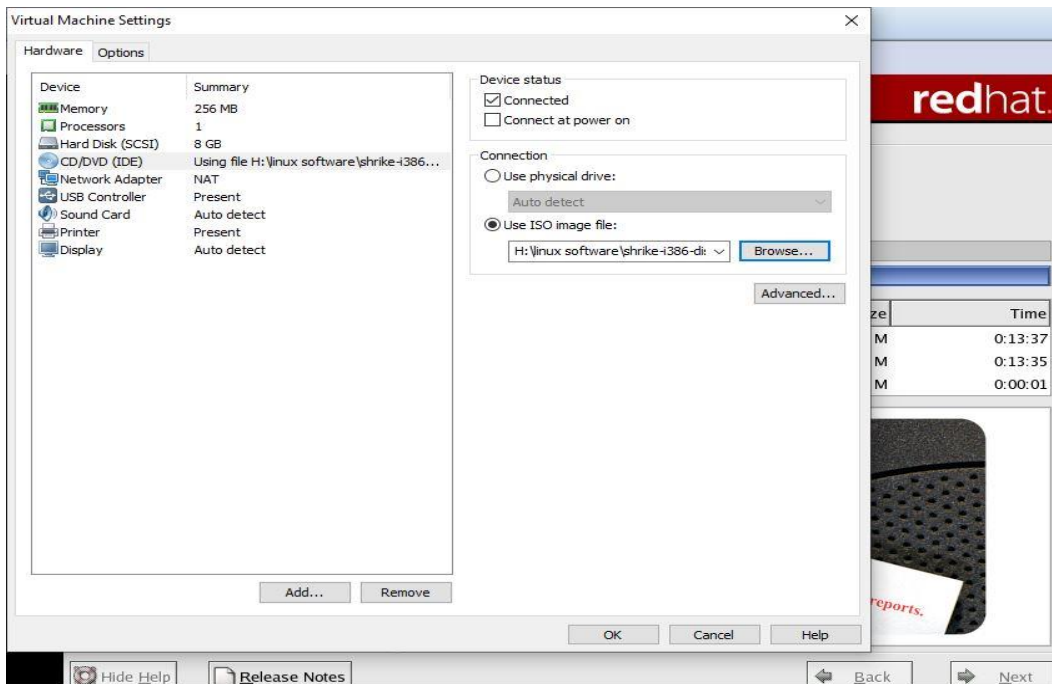
Linux and Shell Scripting



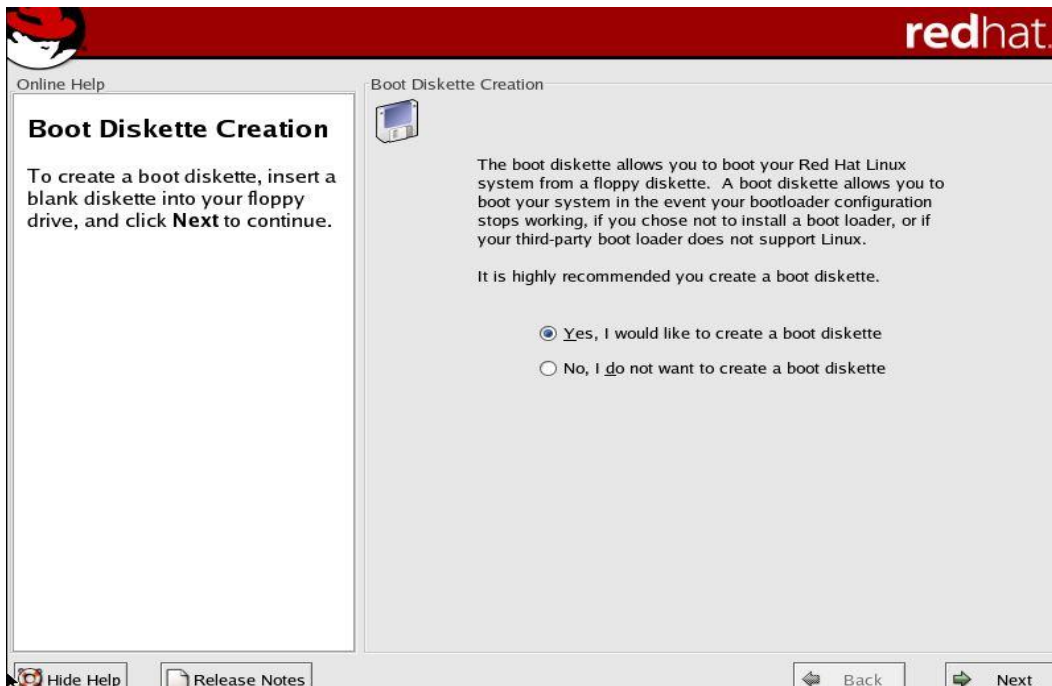
Insert Disc 3



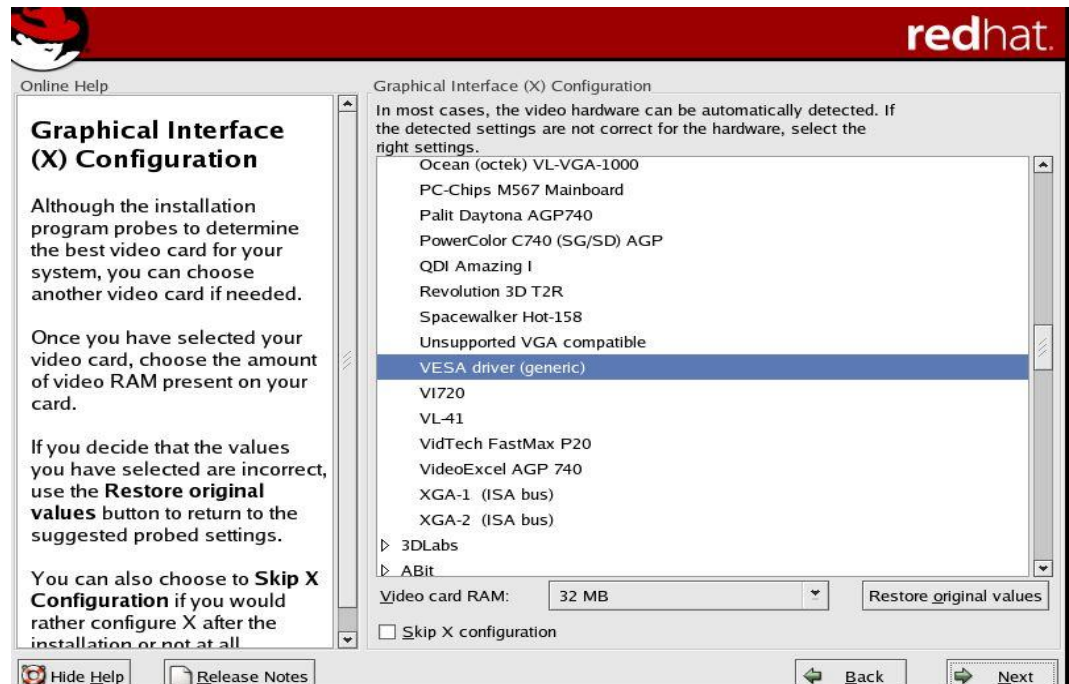
Proper Settings



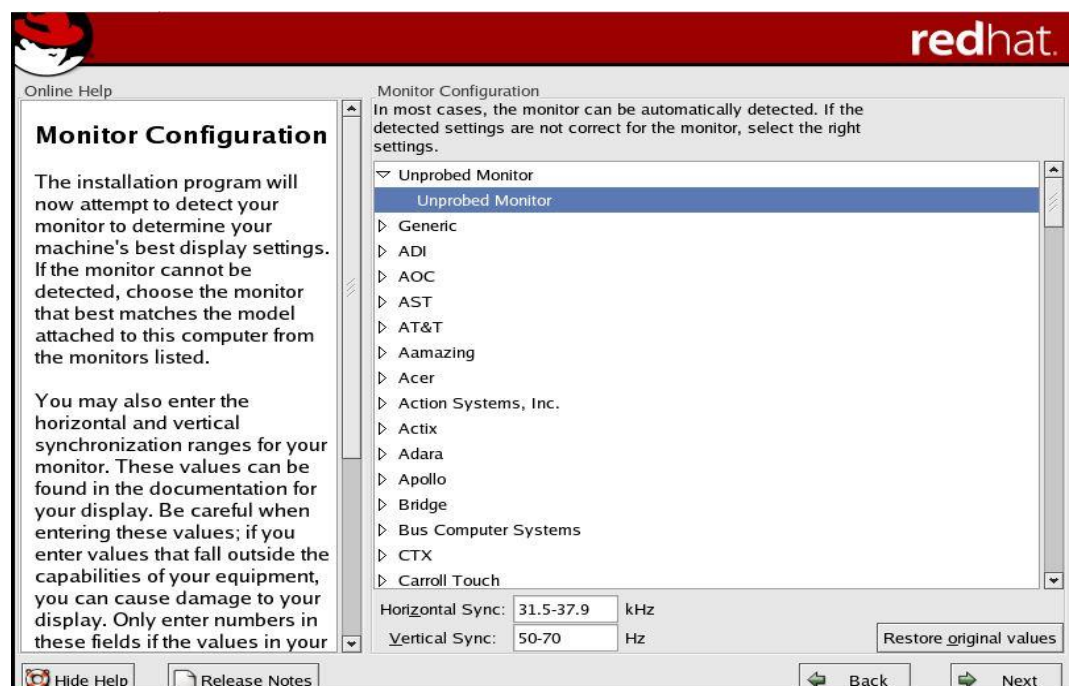
Boot Diskette Creation



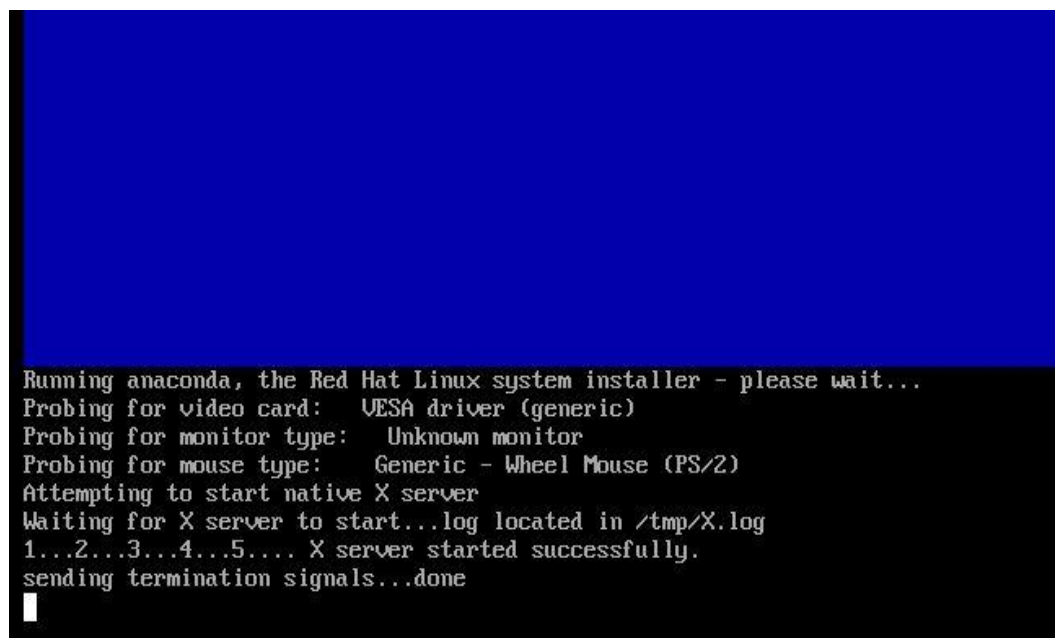
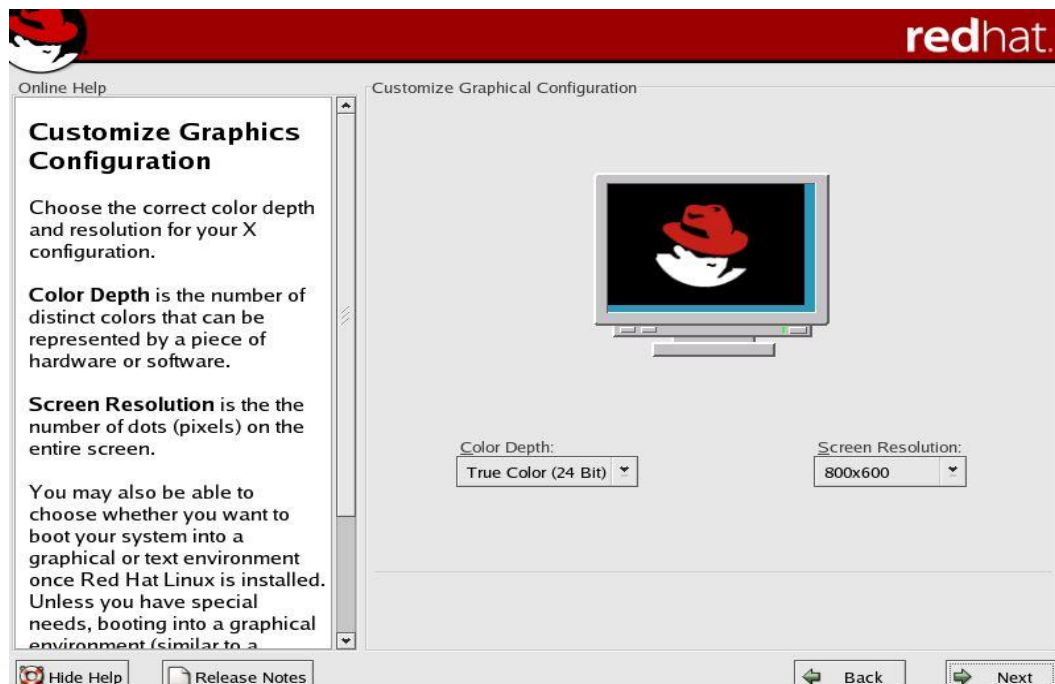
Graphical Interface (X) Configuration



Monitor Configuration



Customize Graphics Configuration



Welcome Screen



Creation of User Account

The image shows the 'User Account' screen of the Red Hat Linux installation process. On the left is a vertical sidebar with a list of steps: 'Welcome', 'User Account' (highlighted with a right-pointing arrow), 'Date and Time', 'Sound Card', 'Red Hat Network', 'Additional CDs', and 'Finish Setup'. The main area has a light gray background. At the top left is a small icon of a user account card. To its right is the text 'User Account' in a large, bold, black font. Below this, a paragraph of text reads: 'It is recommended that you create a personal user account for normal (non-administrative) use. To create a personal account, provide the requested information.' Below the text are four input fields, each preceded by a label: 'Username:', 'Full Name:', 'Password:', and 'Confirm Password:'. Each label is followed by a rectangular text input box. At the bottom right of the main area are two buttons: 'Back' with a left-pointing arrow and 'Forward' with a right-pointing arrow.

Enter Credentials

Welcome

➤ User Account

Date and Time

Sound Card

Red Hat Network

Additional CDs

Finish Setup



User Account

It is recommended that you create a personal user account for normal (non-administrative) use. To create a personal account, provide the requested information.

Username:

Full Name:

Password:

Confirm Password:

◀ Back

▶ Forward

Enter Date and Time

Welcome

User Account

➤ Date and Time

Sound Card

Red Hat Network

Additional CDs

Finish Setup



Date and Time

Please set the date and time for the system.

Date

◀ April ▶ 2021 ▶

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	31	1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	1
2	3	4	5	6	7	8

Time

Current Time : 08:46:10

Hour :

Minute :

Second :

Network Time Protocol

Your computer can synchronize its clock with a remote time server using the Network Time Protocol.

☐ Enable Network Time Protocol

Server:

◀ Back

▶ Forward

Sound Card

Welcome

User Account

Date and Time

▸ Sound Card

Red Hat Network

Additional CDs

Finish Setup



Sound Card

A sound card has been detected on your computer.

Click the "Play test sound" button to hear a sample sound. You should hear a series of three sounds. The first sound will be in the right channel, the second sound will be in the left channel, and the third sound will be in the center.

Vendor: Ensoniq
Model: ES1371 [AudioPCI-97]
Module: es1371

Play test sound

◀ Back

▶ Forward

Registration with Red Hat Network

Welcome

User Account

Date and Time

Sound Card

▸ Red Hat Network

Additional CDs

Finish Setup



Red Hat Network

This step will register your system with a complimentary Demo Account from Red Hat Network so that you can receive the latest software packages directly from Red Hat. Using this tool will allow you to always have the most up-to-date Red Hat Linux system with all the security patches, bug fixes, and software enhancements.

If you purchased this product, you are entitled to a free trial of Red Hat Network Basic Service. To access the free trial, please refer to the product activation card found in the box for detailed instructions.

If you did not purchase this product, visit <http://rhn.redhat.com> for more information or to subscribe to Red Hat Network Basic Service.

☒ Yes, I want to register my system with Red Hat Network.
☐ No, I do not want to register my system.

◀ Back

▶ Forward

Installation of packages from additional CDs



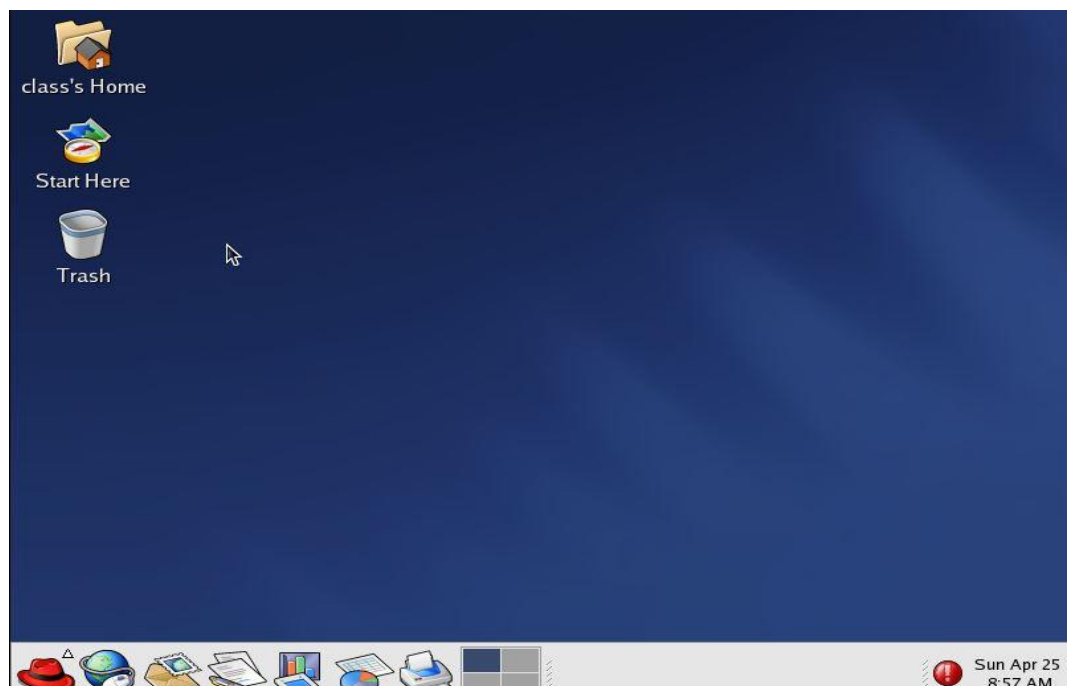
Finish Setup



Logging in



Home Screen



2.4 Moving around the Desktop

- **GNOME:** The default desktop interface of Red Hat Linux 9. GNOME represents a presentation layer that provides a graphical user interface as well as the focused working environment, which enables you to access all your work from one place.
- **KDE:** KDE desktop is included in Red Hat Linux 9 distribution but not installed by default. KDE is a desktop environment for an integrated set of cross-platform applications designed to run on Linux, FreeBSD, Microsoft Windows, Solaris and Mac OS, designed by the KDE Community.

2.5 Components of Desktop

- **Panel:** The panel stretches across the bottom of the desktop. By default, it contains the main menu icon and quick-launch icons for logging out, opening a terminal window, and other common applications and utilities. The panel is highly configurable. You can add and remove buttons that launch applications easily.
- **Workspace:** Workspaces refer to the grouping of windows on your desktop. You can create multiple workspaces, which act like virtual desktops. Workspaces are meant to reduce clutter and make the desktop easier to navigate. Workspaces can be used to organize your work. For example, you could have all your communication windows, such as e-mail and your chat program, on one workspace, and the work you are doing on a different workspace. Your music manager could be on a third workspace.

Summary:

- A boot sequence is the set of operations the computer performs when it is switched on that load an operating system.
- The primary function of BIOS is code program embedded on a chip that recognizes and controls various devices that make up the computer.
- GRUB and LILO are the most popular Linux boot loader.
- The first thing the kernel does is to execute init program.
- There are four types of installation available in Linux: personal desktop, workstation, server and custom.
- There are two ways the disk can be partitioned: manually and automatically.

Keywords:

- **Bootling:** It is a bootstrapping process that starts operating systems when the user turns on a computer system.
- **BIOS:** It refers to the software code run by a computer when first powered on.
- **Partitioning:** It is a means to divide a single hard drive into many logical drives. A partition is a contiguous set of blocks on a drive that are treated as an independent disk.
- **Swap:** Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.
- **GNOME:** The default desktop interface of Red Hat Linux is GNOME.
- **Run-level:** A run-level is a software configuration of the system which allows only a selected group of processes to exist

Self Assessment

1. Which of these executes kernel?
 - A. MBR
 - B. BIOS
 - C. GRUB
 - D. Init

2. The first thing a Kernel does is _____.
 - A. Execute GRUB
 - B. Execute LILO
 - C. Execute init program
 - D. None of the above

3. Which of these tasks are handled by Kernel?
 - A. System Call
 - B. Process Management
 - C. Device Management
 - D. All: System call, process and device management

4. While installation of Red Hat Linux in the system, it asks for _____.
 - A. Language Selection
 - B. Keyboard Configuration
 - C. Mouse Configuration
 - D. All: Language selection, keyboard and mouse configuration

5. In partition field, i.e., SIZE, the measurement unit is _____.
 - A. TB
 - B. MB
 - C. GB
 - D. KB

6. Which of these defines the runlevels?
 - A. 0-6
 - B. 1-7
 - C. 2-8
 - D. 3-9

7. Which of these partitions is used to support virtual memory?
 - A. /
 - B. /var
 - C. /boot
 - D. swap

8. Which of these is a Red Hat Linux installer?
- A. Anaconda
 - B. GRUB
 - C. LILO
 - D. Emulator
9. Which of these programs allows us to partition the disk?
- A. Anaconda
 - B. GRUB
 - C. Disk Druid
 - D. Disk Help
10. MBR is executed by _____ .
- A. BIOS
 - B. GRUB
 - C. Kernel
 - D. Init
11. In which mode, it is possible to install and upgrade Red Hat Linux?
- A. Graphical
 - B. Text
 - C. Both graphical and text
 - D. None of the above
12. What does BIOS mean?
- A. Basic Input/ Output Service
 - B. Basic Input/ Output System
 - C. Buffer Input/ Output System
 - D. Buffer Input/ Output Service
13. MBR is executed by _____ .
- A. BIOS
 - B. GRUB
 - C. Kernel
 - D. Init
14. In which mode, it is possible to install and upgrade Red Hat Linux?
- A. Graphical
 - B. Text
 - C. Both graphical and text
 - D. None of the above
15. What type can be installed in Red Hat Linux?

- A. Personal Desktop
- B. Server
- C. Workstation
- D. All: personal desktop, server and workstation

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. D | 4. D | 5. B |
| 6. A | 7. D | 8. A | 9. C | 10. A |
| 11. C | 12. B | 13. A | 14. C | 15. D |

Review Questions:

1. What is booting? Explain the booting sequence in detail.
2. What is a kernel? Explain the tasks of a kernel in detail.
3. What is a partition? Write the partition fields. What is the recommended partition scheme?
4. What is a file system? Explain its types in detail.
5. What is a run-level? Explain about the run-level of in it.



Further Readings

Mark G. Sobell, A Practical Guide to Fedora and Red Hat Enterprise Linux, Fifth Edition, Pearson Education, Inc.



Web Links

<https://www.educba.com/install-linux/>

<https://phoenixnap.com/kb/linux-create-partition>